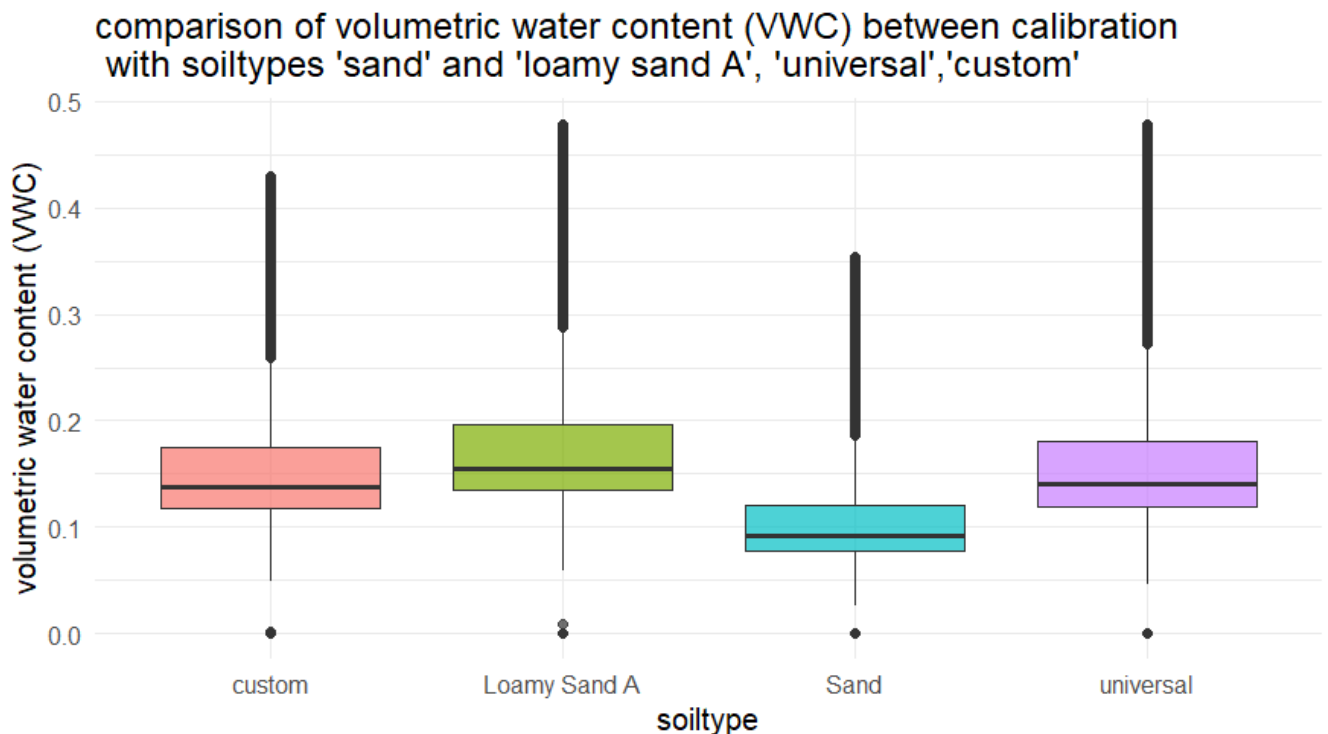


Is a custom laboratory calibration necessary?

The plot below shows our measured VWC values, calibrated with different calibration approaches



- Differences are all significant (as expected, systematically different)
- Quantiles of differences in the following order
 - Sand- loamy Sand A (diffs)
 - Universal- custom (diffs2)
 - Universal- sand (diffs 3)
 - Custom density 1- sand (diffs 4)
 - Custom density 1.5- universal (diffs 6)
 - Custom density 1.7- universal (diffs/)
- For universal-custom, the differences are mostly below the accuracy of the soil moisture sensor (+/- 1%)
- Problem: values for custom were calculated with a tool, developed for TMS3- TMS4 could have different initial values (To get the parameters I used the soil analysis from Landcare from Bull Pass East) “Universal” was developed for TMS4

```

> quantile(diffs, na.rm = T)
      0%      25%      50%      75%     100%
-0.12508593 -0.07467849 -0.06318347 -0.05707957  0.00000000
> quantile(diffs2, na.rm = T)
      0%      25%      50%      75%     100%
-0.003102626  0.001548111  0.003070809  0.006422635  0.047801675
> quantile(diffs3, na.rm = T)
      0%      25%      50%      75%     100%
0.00000000  0.04227150  0.04817367  0.05955600  0.12452924
> quantile(diffs4, na.rm = T)
      0%      25%      50%      75%     100%
0.00000000  0.04072348  0.04510190  0.05313572  0.07807969
>

```

```

> quantile(diffs6, na.rm = T)
      0%      25%      50%      75%     100%
0.000000000  0.009352612  0.011028852  0.014731054  0.060924606
> quantile(diffs7, na.rm = T)
      0%      25%      50%      75%     100%
0.000000000  0.007122923  0.008755166  0.012357748  0.057179437
>

```

```

diffs <- sand_vwc_agg_long$value - loamySandA_vwc_agg_long$value
hist(diffs, main = "distribution of differences", xlab = "difference (Sand - Loamy
boxplot(diffs)
quantile(diffs, na.rm = T)

```

Do I need custom calibration?

We use TMS4 sensors. There are calibration curves available for TMS3 sensors (soil type dependent, but are also used for TMS4. For TMS4 there is a calibration curve for a “universal” soil type available. To find out whether I should do a custom calibration, I compared differences of the, with different curves calibrated, data. For custom calibration, the soils bulk density is needed. This is not available for antarctica, because you can not get an undisturbed soil sample from those soils. Lab measurements from landcare indicates, that our soil type should be somewhere between Sand and loamy sand A calibration curves. A check with the given lab results, used to calculate a custom curve (with the TMS3 tool) supports this. Because bulk density remains unknown, but is needed, this was done for 1.0, 1.5 and 1.7 g/cm³ density, which reflects the range of bulk density for unknown soil type. Comparing those three shows, that the bulk density will have an effect on the resulting calibration curve(, while different sensors won’t have a big influence.) Comparing these custom calibration curves to the universal curve (TMS4) shows, that the differences between the results are ranging around 1%. As 1% is also the accuracy of the soil moisture sensor, this is expected to not have a big influence on the results. Another thing you can see (boxplots) is, that the higher moisture measurements will be affected more than the dry condition measurements. Soil for calibration in the lab is available from the DryValleys, but not specifically from Taylor valley. The bulk density is expected to be heterogenous throughout the valley (differences at least around +- 20% along the valley, see Megans Email). Because those factors will also lead to not further specifiabe uncertainties, we decided to use the

universal soil calibration curve, designed for TMS4, to assure better compatibility to other studies and to avoid inaccessible uncertainties.